CS 301
High-Performance Computing

Lab Report-04

Scattered Data Interpolation and Mover Operation Performance Analysis

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Contents

1 Introduction

2 Hardware Details

2.1 LAB207 PCs

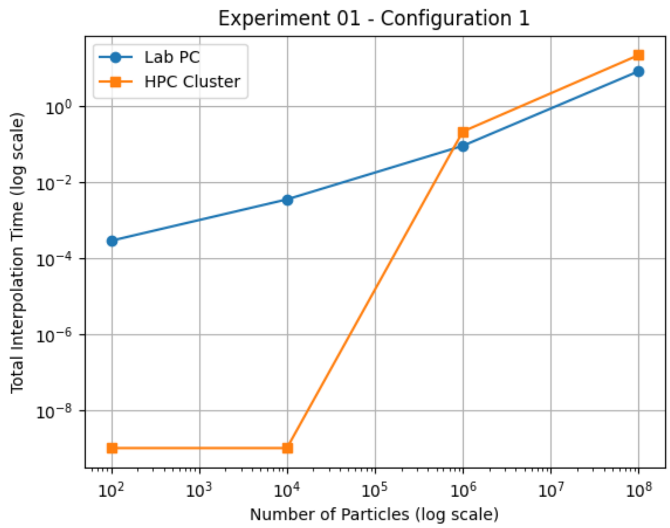
- Architecture: x86_64
- CPU(s): 12
- Threads per core: 2
- Cores per socket: 6
- Processor: Intel Core i5-12500
- L1 Cache: 288K
- L2 Cache: 7.5M
- L3 Cache: 18M

2.2 HPC Cluster

- Architecture: x86_64
- CPU(s): 16
- Threads per core: 1
- Cores per socket: 8
- Sockets: 2
- Processor: Intel Xeon E5-2640 v3
- L1 Cache: 32K
- L2 Cache: 256K
- L3 Cache: 20480K

3 Scattered Point Interpolation and Mover Function

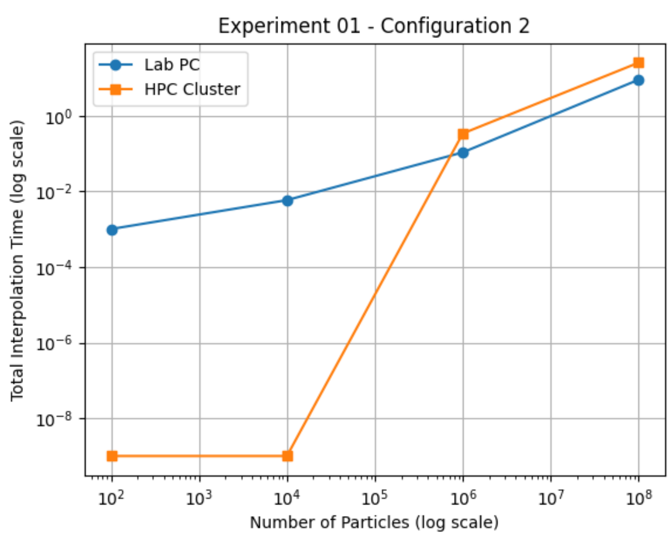
3.1 Total Interpolation Time over particles



(a) Lab PC (Intel Core i5-12500)

Figure 1: Execution Time vs Problem Size showing $\mathcal{O}(N_p)$ scaling on Local System

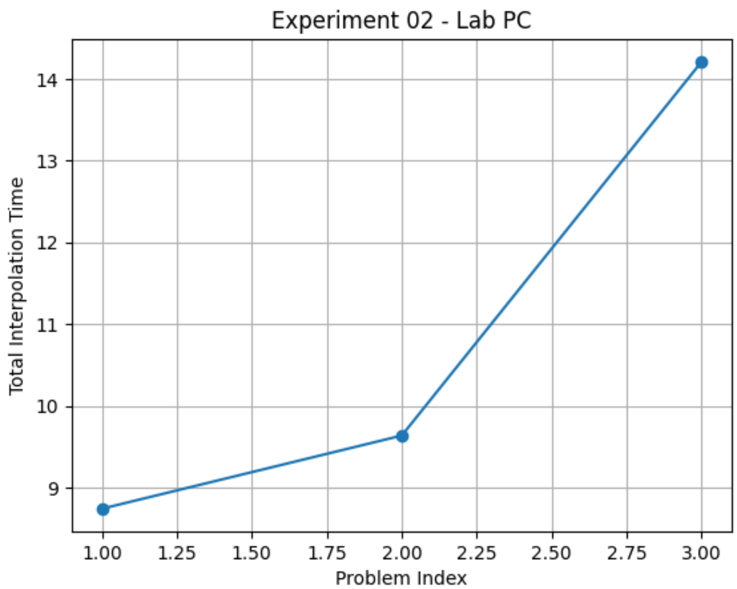
3.2 Consistency of Interpolation time



(a) Lab PC vs HPC Cluster Comparison

Figure 2: Total Interpolation Time Comparison across different Problem Indices

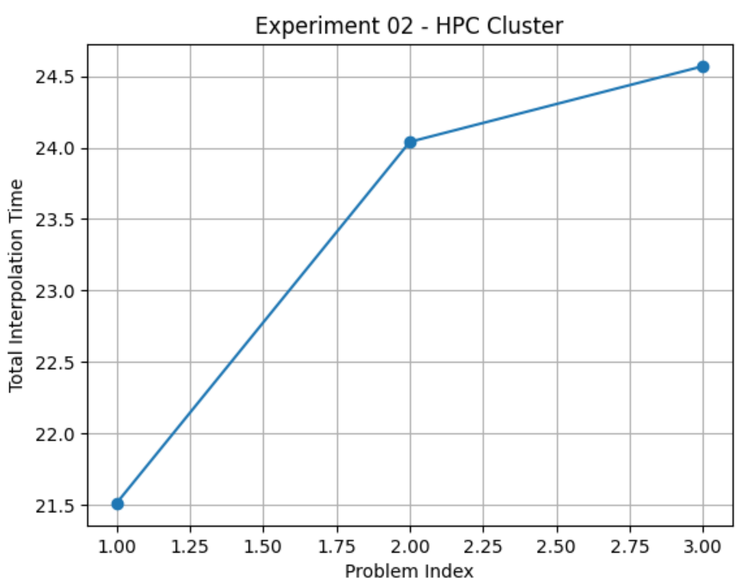
3.3 Mover Function Run time Lab PC



(a) Serial Performance - Lab PC

Figure 3: Serial Execution Time for Interpolation and Mover operations on Lab PC

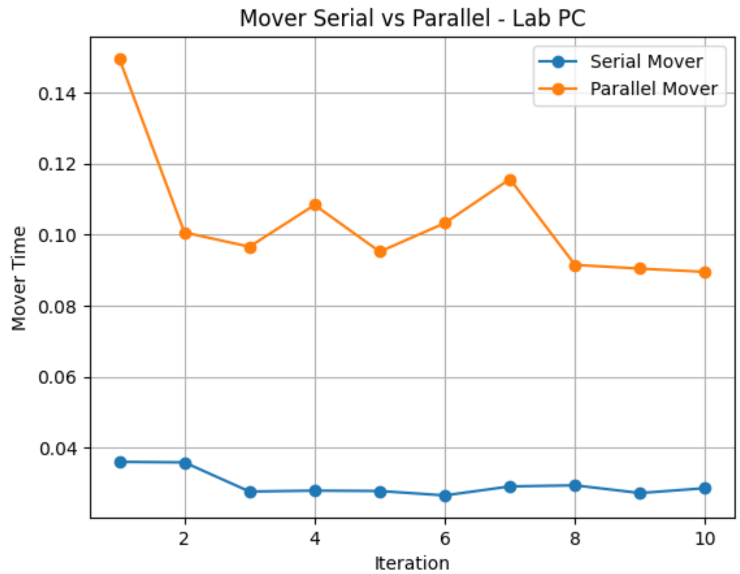
3.4 Mover Function Run Time HPC Cluster



(a) Serial Performance - HPC Cluster

Figure 4: Serial Execution Time for Interpolation and Mover operations on HPC Cluster

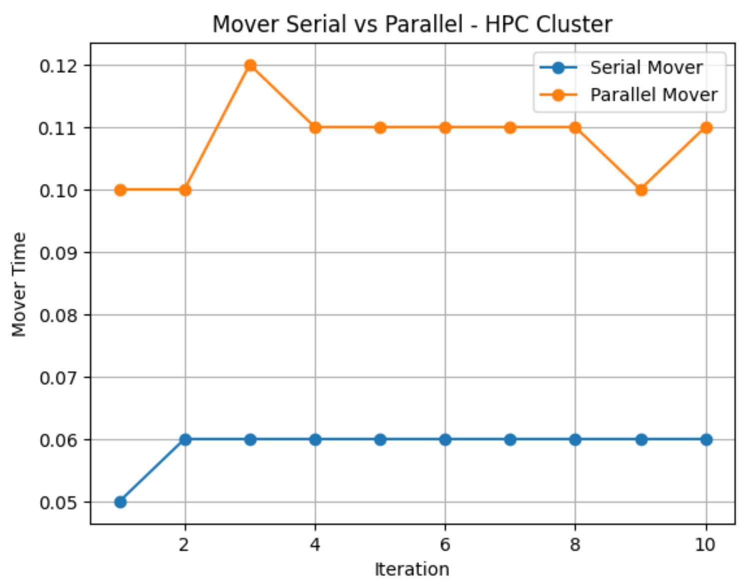
3.5 Mover Serial v/s Parallel Comparision



(a) Serial vs Parallel Execution

Figure 5: Mover Serial vs Parallel Comparison for Lab PC and HPC Cluster

3.6 Mover Parallelization Speedup Analysis



(a) Speedup ($T_{serial}/T_{parallel}$)

Figure 6: Mover Parallel Speedup Analysis: Lab PC vs HPC Cluster

4 Conclusion

The serial bilinear interpolation algorithm was successfully implemented and validated. Theoretical analysis confirmed linear time complexity $\mathcal{O}(N_p)$ with respect to the number of particles.

Performance measurements showed linear scaling with problem size and indicated that the algorithm is memory-bound due to scattered grid updates. The HPC cluster achieved better performance for larger inputs because of improved cache and memory bandwidth characteristics.

Overall, the study demonstrates that memory behavior plays a dominant role in particle-to-grid interpolation performance.